



## Experiment No.: 4

**Title:** Use function types, scope, and closures; apply try-catch; build a password strength checker using palindrome then display strong or weak password and generate a html and js code.

**Software/Tools Required:** HTML5, CSS3, JavaScript, VS Code, Google Chrome

### Index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Password Strength Checker</title>
<style>
*{
  margin:0;
  padding:0;
  box-sizing:border-box;
  font-family:'Lucida Console', 'Monaco', monospace;
}
body{
  height:100vh;
  display:flex;
  justify-content:center;
  align-items:center;
  background:linear-gradient(135deg,#89949810,#84b2da);
}
.container{
  width:400px;
  background:white;
  border-radius:18px;
  padding:35px;
  text-align:center;
  box-shadow:
    0 15px 35px rgba(0,0,0,0.2);
}
.icon{
  font-size:55px;
  margin-bottom:10px;
}
h2{
```

---



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```
color:#333;
margin-bottom:8px;
}
p.subtitle{
color:#777;
margin-bottom:25px;
font-size:14px;
}
.input-box{
position:relative;
}
input{
width:100%;
padding:14px 16px;
border:2px solid #ddd;
border-radius:10px;
outline:none;
font-size:16px;
transition:.3s;
}
input:focus{
border-color:#33a060;
}
button{
width:100%;
margin-top:18px;
padding:14px;
border:none;
border-radius:10px;
background:#667eea;
color:white;
font-size:16px;
font-weight:600;
cursor:pointer;
transition:.3s;
}
button:hover{
background:#4f61d8;
transform:translateY(-2px);
}
#result{
margin-top:22px;
padding:14px;
border-radius:10px;
```



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Experiment-4

```
background:#f7f7f7;
font-size:17px;
font-weight:600;
min-height:55px;
display:flex;
justify-content:center;
align-items:center;
}
#count{
margin-top:15px;
color:#555;
font-size:14px;
}
.footer{
margin-top:20px;
font-size:12px;
color:#999;
}
</style>
</head>
<body>
<div class="container">
<div class="icon">🔒 </div>
<h2>Password Strength Checker</h2>
<p class="subtitle">
A palindrome password is considered <b>Weak</b>.
Any other password is considered <b>Strong</b>.
</p>
<div class="input-box">
<input
type="password"
id="password"
placeholder="Enter your password">
</div>
<button onclick="checkPassword()">
Check Password
</button>
<div id="result">
Result will appear here...
</div>
<div id="count">
</div>
</div>
<script src="script.js"></script>
```



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</body>

</html>

**Script.js:**

```
let appName = "Password Strength Checker";
function passwordCounter(){
    let count = 0;
    return function(){
        count++;
        return count;
    };
}
const counter = passwordCounter();
function isPalindrome(password){
    let reverse = password.split("").reverse().join("");
    return password === reverse;
}
const displayResult = function(message, color){
    const result = document.getElementById("result");
    result.innerHTML = message;
    result.style.color = color;
    if(color === "green"){
        result.style.background = "#e8f9ee";
        result.style.border = "2px solid #28a745";
    }
    else if(color === "red"){
        result.style.background = "#ffe8e8";
        result.style.border = "2px solid #dc3545";
    }
    else{
        result.style.background = "#fff4d6";
        result.style.border = "2px solid #ff9800";
    }
};
const updateCount = ()=>{
    document.getElementById("count").innerHTML =
    "Passwords Checked : " + counter();
};
function checkPassword(){
    try{
        let password = document.getElementById("password").value;
        if(password.trim() === ""){
            throw "Password cannot be empty.";
        }
        if(password.length < 4){
```



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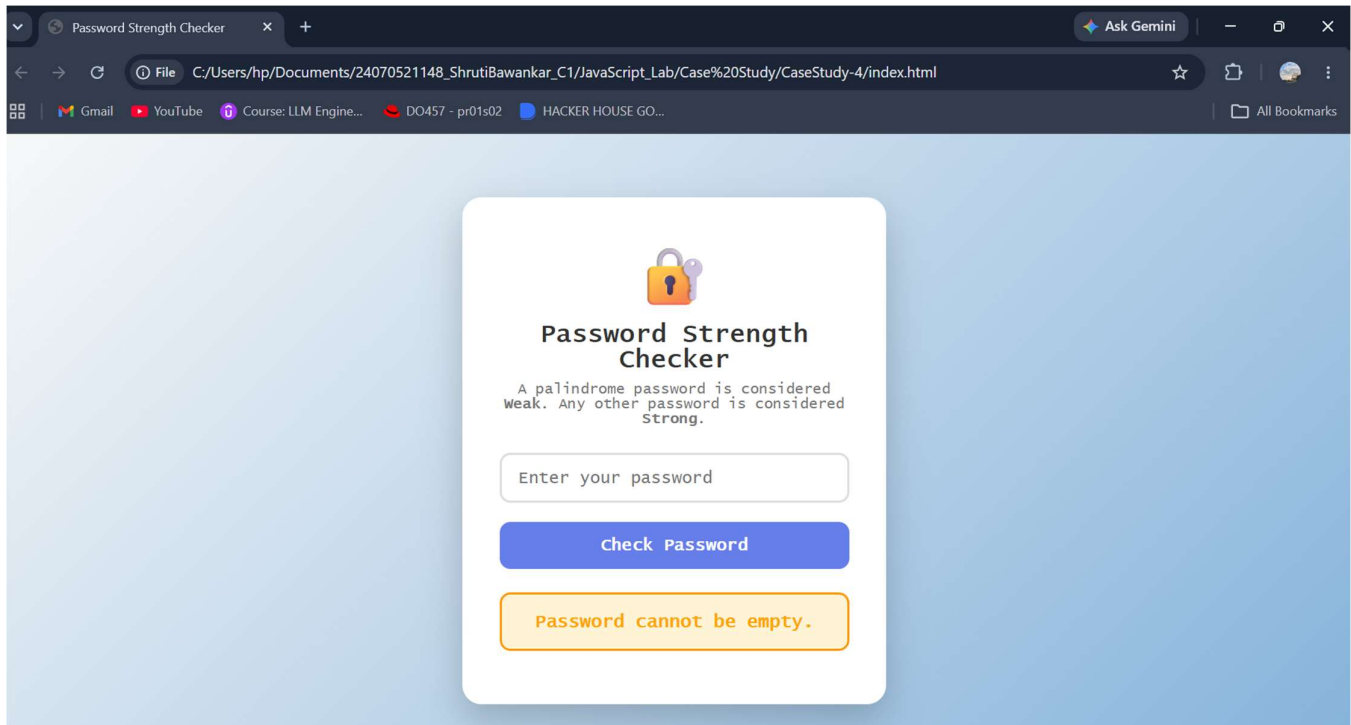
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## Experiment-4

```
        throw "Password must contain at least 4 characters.";
    }
    if(isPalindrome(password)){
        displayResult(
            "Weak Password (Palindrome Password)",
            "red"
        );
    }
    else{
        displayResult(
            "Strong Password (Not a Palindrome)",
            "green"
        );
    }
    updateCount();
}
catch(error){
    displayResult(error,"orange");
}
}
```

### Output:





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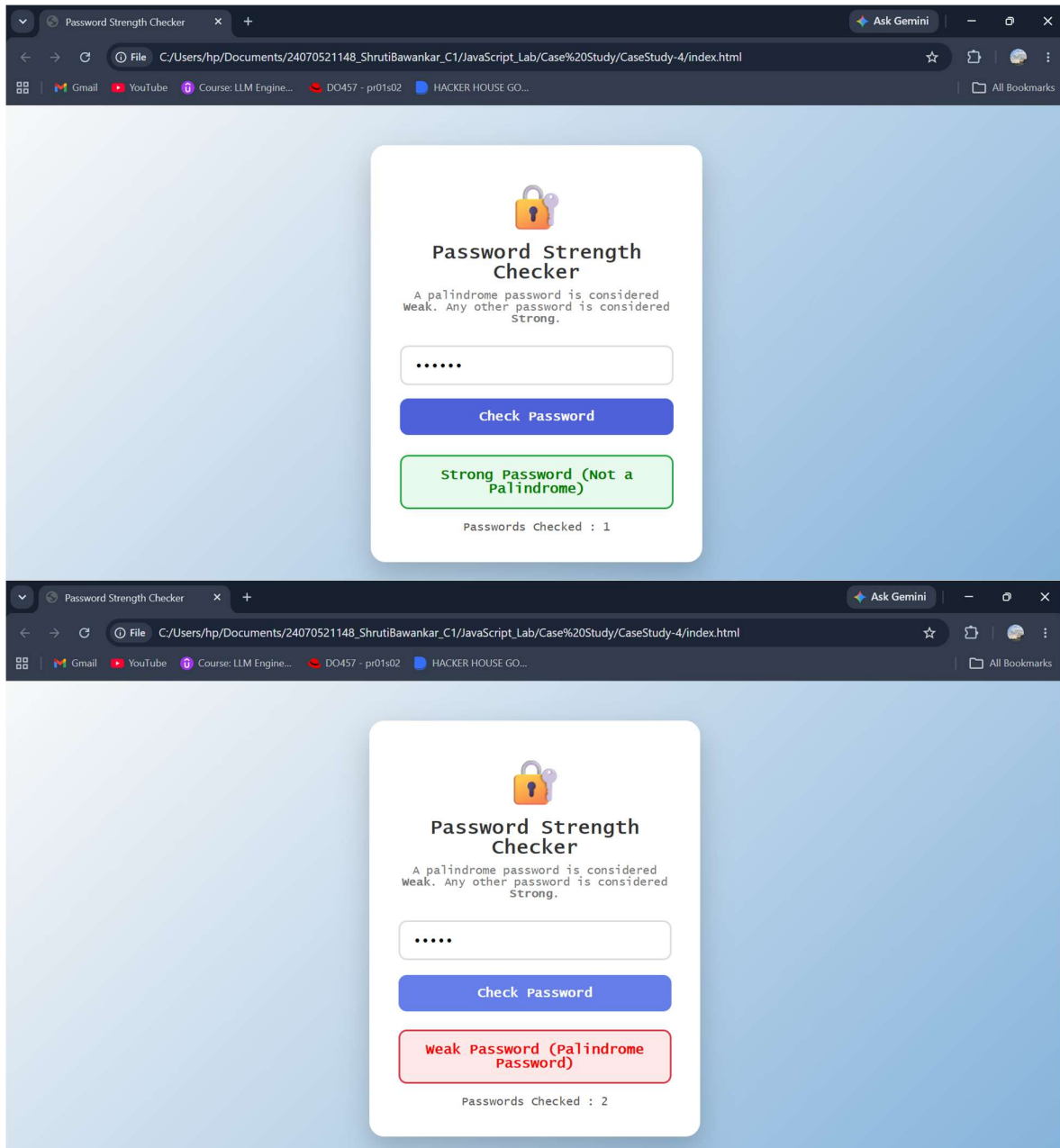
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## Experiment-4



**Result:** The password checker successfully identifies whether a password is a palindrome and classifies it as **Weak** or **Strong** while handling invalid inputs using try...catch.

**Conclusion:** The program effectively demonstrates JavaScript function types, scope, closures, and exception handling through a simple and interactive password validation application.